

Puzzles to Puzzle You (Remastered)

Welcome to the sixth iteration of *Puzzles to Puzzle You (Remastered)* by Rani Rachavelpula, in which I will discuss the previous puzzle (Pythagorean Tuning) and pose the next one.

Puzzle #5: Pythagorean Tuning

Consider the Pythagorean tuning system, where tones are generated by placing points on the unit circle at fractional positions $n\theta$ for $n = 0, 1, 2, \dots, 12$. Let $\theta = \frac{7}{12} + \delta$ be an irrational number close to $\frac{7}{12}$.

1. Determine how many distinct intervals (arc lengths) exist between adjacent tones in this sequence.

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2. What are the possible numbers of distinct interval lengths?

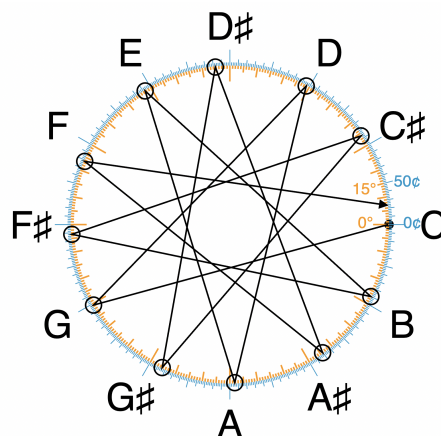
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3. What does this say about the structure of the tuning system

- Adjacent tones can occupy three possible tuning discrepancies. The uneven spacing of the Pythagorean tuning system is in contrast to the 12-tone equal temperament, perhaps explaining why some intervals (like major thirds) are noticeably out of tune in Pythagorean tuning.

Shout Out No one embodies the ethos of this newsletter more than my friend and comrade Wei. Though Wei has not solved any puzzles yet, he expanded my sense of mathematical community with his summer Peter Lax reading group. A few weeks ago, the inimitable Peter Lax — once dubbed ‘the most versatile mathematician of his generation’ — passed away at the age of 99. Among his many accomplishments, Lax was the first applied mathematician to receive the Abel Prize in 2005. To

be a member of Wei’s reading group has been an honor which has gifted me both knowledge and friendship. I hope this newsletter can facilitate a similar sense of enlightenment and fellow-feeling.



Pythagorean Circle of Fifths

My Solution To solve the first part of this puzzle, all we need to do is determine the first 13 terms of the sequence $\{n\theta \bmod 1\}$ for $\theta = \frac{7}{12} + \delta$, where δ is small irrational number.

I chose $\theta = \frac{7}{12} + \frac{\pi}{1000}$. The following table summarizes the first 13 terms of the sequence:

n	$n\theta$	$n\theta \bmod 1$
0	0	0
1	$\frac{7}{12} + \frac{\pi}{1000}$	$\frac{7}{12} + \frac{\pi}{1000}$
2	$\frac{14}{12} + \frac{2\pi}{1000}$	$\frac{2}{12} + \frac{2\pi}{1000}$
3	$\frac{21}{12} + \frac{3\pi}{1000}$	$\frac{9}{12} + \frac{3\pi}{1000}$
4	$\frac{28}{12} + \frac{4\pi}{1000}$	$\frac{4}{12} + \frac{4\pi}{1000}$
5	$\frac{35}{12} + \frac{5\pi}{1000}$	$\frac{11}{12} + \frac{5\pi}{1000}$
6	$\frac{42}{12} + \frac{6\pi}{1000}$	$\frac{6}{12} + \frac{6\pi}{1000}$
7	$\frac{49}{12} + \frac{7\pi}{1000}$	$\frac{1}{12} + \frac{7\pi}{1000}$
8	$\frac{56}{12} + \frac{8\pi}{1000}$	$\frac{8}{12} + \frac{8\pi}{1000}$
9	$\frac{63}{12} + \frac{9\pi}{1000}$	$\frac{3}{12} + \frac{9\pi}{1000}$
10	$\frac{70}{12} + \frac{10\pi}{1000}$	$\frac{10}{12} + \frac{10\pi}{1000}$
11	$\frac{77}{12} + \frac{11\pi}{1000}$	$\frac{5}{12} + \frac{11\pi}{1000}$
12	$\frac{84}{12} + \frac{12\pi}{1000}$	$\frac{0}{12} + \frac{12\pi}{1000}$

Now, we order the values of $n \bmod 1$ according to the usual ordering on $\mathbb{R}$ and we calculate the difference between adjacent pairs.

$$\begin{array}{lcl}
 1. & 0 & \\
 2. & \frac{0}{12} + \frac{12\pi}{1000} & \left\{ \begin{array}{l} \frac{12\pi}{1000} \\ \frac{1}{12} - \frac{5\pi}{1000} \end{array} \right. \\
 3. & \frac{1}{12} + \frac{7\pi}{1000} & \left\{ \begin{array}{l} \frac{1}{12} - \frac{5\pi}{1000} \\ \frac{1}{12} + \frac{7\pi}{1000} \end{array} \right. \\
 4. & \frac{2}{12} + \frac{2\pi}{1000} & \left\{ \begin{array}{l} \frac{1}{12} - \frac{5\pi}{1000} \\ \frac{1}{12} + \frac{7\pi}{1000} \end{array} \right. \\
 5. & \frac{3}{12} + \frac{9\pi}{1000} & \left\{ \begin{array}{l} \frac{1}{12} - \frac{5\pi}{1000} \\ \frac{1}{12} + \frac{7\pi}{1000} \end{array} \right. \\
 6. & \frac{4}{12} + \frac{4\pi}{1000} & \left\{ \begin{array}{l} \frac{1}{12} - \frac{5\pi}{1000} \\ \frac{1}{12} + \frac{7\pi}{1000} \end{array} \right. \\
 7. & \frac{5}{12} + \frac{11\pi}{1000} & \left\{ \begin{array}{l} \frac{1}{12} - \frac{5\pi}{1000} \\ \frac{1}{12} + \frac{7\pi}{1000} \end{array} \right. \\
 8. & \frac{6}{12} + \frac{6\pi}{1000} & \left\{ \begin{array}{l} \frac{1}{12} - \frac{5\pi}{1000} \\ \frac{1}{12} + \frac{7\pi}{1000} \end{array} \right. \\
 9. & \frac{7}{12} + \frac{\pi}{1000} & \left\{ \begin{array}{l} \frac{1}{12} - \frac{5\pi}{1000} \\ \frac{1}{12} + \frac{7\pi}{1000} \end{array} \right. \\
 10. & \frac{8}{12} + \frac{8\pi}{1000} & \left\{ \begin{array}{l} \frac{1}{12} - \frac{5\pi}{1000} \\ \frac{1}{12} + \frac{7\pi}{1000} \end{array} \right. \\
 11. & \frac{9}{12} + \frac{3\pi}{1000} & \left\{ \begin{array}{l} \frac{1}{12} - \frac{5\pi}{1000} \\ \frac{1}{12} + \frac{7\pi}{1000} \end{array} \right. \\
 12. & \frac{10}{12} + \frac{10\pi}{1000} & \left\{ \begin{array}{l} \frac{1}{12} - \frac{5\pi}{1000} \\ \frac{1}{12} + \frac{7\pi}{1000} \end{array} \right. \\
 13. & \frac{11}{12} + \frac{5\pi}{1000} & \left\{ \begin{array}{l} \frac{1}{12} - \frac{5\pi}{1000} \\ \frac{1}{12} + \frac{7\pi}{1000} \end{array} \right. \\
 14. & 1 \equiv 0 &
 \end{array}$$

From this series of calculations, we see that there are three distinct intervals (arc lengths) between adjacent tones in this sequence.

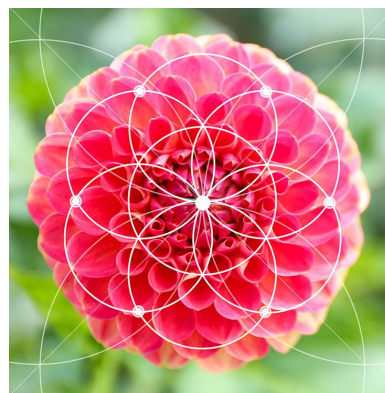
$$\begin{array}{l}
 1. \quad \frac{12\pi}{1000} \\
 2. \quad \frac{1}{12} - \frac{5\pi}{1000} \\
 3. \quad \frac{1}{12} + \frac{7\pi}{1000}
 \end{array}$$

The second part of the puzzle generalizes the first and asks how many distinct intervals can be generated if we do not stop the sequence after 13 terms, but continue indefinitely.

Incredibly, no matter how many terms of the sequence we take, adjacent pairs can only differ by three possible values. This result follows directly from the Steinhaus conjecture aka the three-gap theorem.

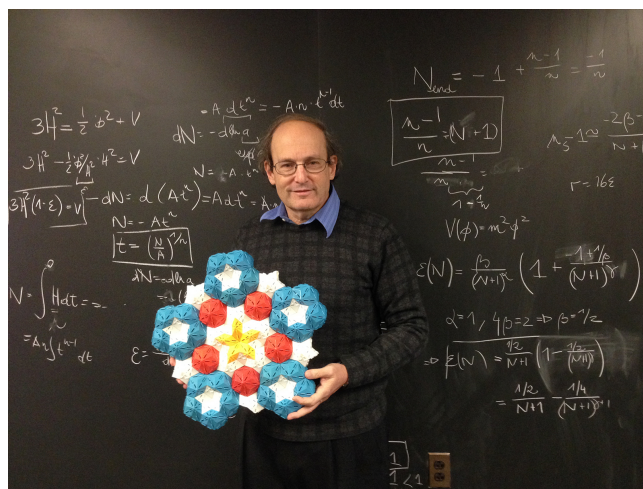
THM (Steinhaus Conjecture) If one places n points on a circle, at angles of $\theta, 2\theta, 3\theta, \dots$ from the starting point, then there will be at most three distinct distances between pairs of adjacent points. When there are three distances, the largest of the three always equals the sum of the other two.

I will not prove the Steinhaus Conjecture. Instead, I want to highlight two beautiful real-life applications. My favorite is in the study of plant growth, or phyllotaxis. Scientists have found that successive leaves on many plant stems are separated by the golden angle—about 137.5° . This pattern is believed to optimize sunlight exposure.



Fibonacci Flower

Another application is to quasi-crystals — solids with ordered atomic structures that lack periodic repetition. Their atomic arrangements are modeled by ‘cut-and-project sets’ using irrational slopes, generating non-repeating but uniformly distributed point sets.

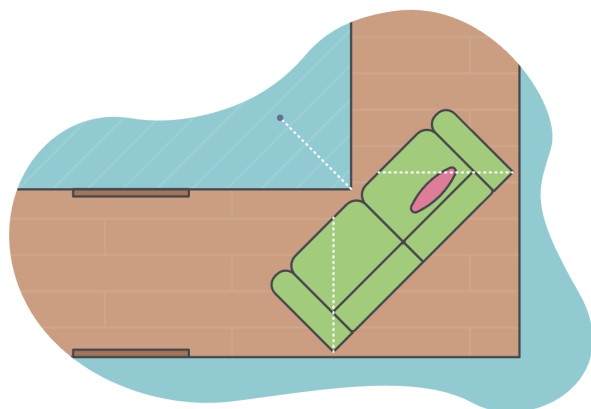


Paul Steinhardt, co-discover of the first known natural quasicrystal, with a constructible model of a quasi-crystal.

Puzzle #6: Moving Sofa Problem (Variation)

You have a hallway of width 1 that makes a 90° turn (an L-shaped corridor). Your sofa is *pill-shaped*. Specifically, it is a rectangle of length L and width 1. Attached to both ends of the rectangle are semicircles of radius $\frac{1}{2}$.

1. What is the maximal length L of the rectangle part so that this pill-shaped sofa can be moved around the corner without colliding with the hallway walls?



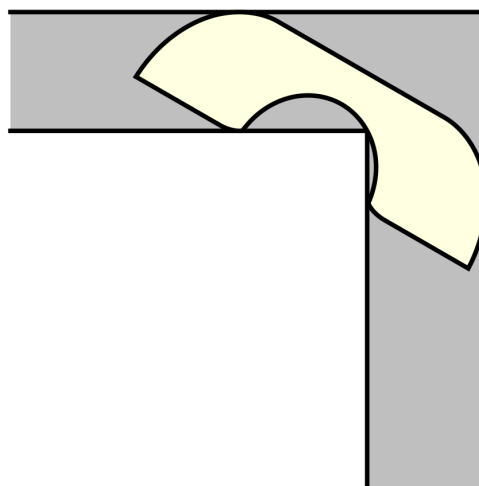
*Moving Sofa Problem for a
Rectangular-Shaped Sofa*

The original Moving Sofa Problem was first formally posed by Leo Mosser in 1966.

Original Moving Sofa Problem What is the shape of largest area in the plane that can be moved around a right-angled corner in a two-dimensional hallway of width 1?

In 1992 American mathematician Joseph Gerver proposed a solution involving a telephone shaped sofa with area ≈ 2.2195 . For over twenty years, Gerver's sofa was the leading solution though a proof evaded the mathematical community.

In 2024, Korean mathematician Jineon Baek proved the optimality of Gerver's sofa. The paper in which Baek presents his landmark proof is over 100 pages long. Unlike previous efforts that utilized extensive computational resources to approximate upper bounds, Baek's utilized analytical methods from the calculus of variations which provided a more elegant and potentially more generalizable solution.

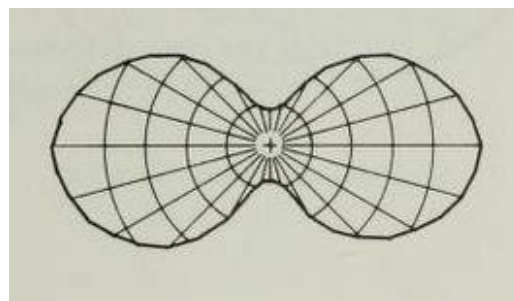


Gerver's Sofa

The Sofa Moving Problem is one of many so-called 'furniture moving problem.' Since its original proposal by Mosser, mathematicians have come up with several variations. One notable variation is the The Conway Car Problem, so-named after the 'magical genius' and Joseph Conway.

Conway's Car Problem What is the longest rigid body of unit width (like a car) that can drive around a right-angled corner of a hallway — without reversing?

It's important to note the difference between Conway's problem and the Sofa problem. In particular, Conway's problem asks to optimize the length of a car (not the area). It further adds the constraint that only forward motion is allowed.



Muruyama's Car Solution

The leading solution for Conway's Car Problem came from Muruyama in 1973. The shape of Muruyama's solution has been dubbed 'the bikini' for obvious reasons.